Supplement: methods, sensitivities and audit trail

For *Older than their fossils? What the Ordovician–Silurian record can test about the maxima that divide land-plant molecular clocks*.

Barry W. Brook*[1], Claude Opus 5.5 (Anthropic) and OpenAI Codex[2]

Internal research draft from a living repository. Release and submission are Barry W. Brook's decision.

1. University of Tasmania, Hobart, Tasmania, Australia
2. AI research collaborators; roles are stated in "Roles of AI systems and humans"

Corresponding author: Barry W. Brook

This supplement holds the machinery behind the main text. Repository paths are given so that every number can be traced to the table or script that produced it. Internal labels appear here because the repository uses them: **M1**, **M2** and **M3** are the 515.5, 459.1 and 431.2 Ma maxima; **E1** trilete spores, **E2** cryptospores accepted as embryophytic under a named reading, **E3** polysporangiate bodies, and **E5** crown vascular-plant members. **B0**, **B1** and **B2** are the body-record categories: any land-plant body (any accepted macro- or mesofossil; canonically including the Katian Oman sporangial fragments, S5.1), polysporangiophyte body and crown vascular-plant member. `manuscript/estimands.md` states what each analysis bounds and which phrasings are not allowed.

S1. Readings, observations and units

Named readings. Disputed identifications, affinities and ages are carried as alternatives, not resolved by fiat. The *main* reading accepts what the cited sources accept. The *inclusive* reading adds records that a named source accepts against others, and the *outlier* reading admits further records that are explicitly contested. A wider reading requires a record-specific acceptance; it does not promote every unresolved grain or axis to a detection. The E2 rule separates spore morphology and producer evidence from the embryophyte affinity accepted by each reading (`data/units/e2-unit-rule.md`). Rejected, reassigned and disputed records are listed in `docs/rejected-records.md`.

Observations. Within each frame, the same reporting rule applies to older and reference material. An explicit non-detection is scoped to the material actually examined. A missing report becomes a zero only after a full-category reporting audit. Unresolved identification, missing links and unreported effort remain distinct from absence.

Ages. Stage bases follow ICS v2024/12 (`data/timescale.csv`). Source-specific age statements and their numerical conversions are retained separately. Literature before 2024 mostly uses GTS2012 Early Devonian boundaries, which differ from ours by up to 3 Myr. Uniform within-range ages and shared within-range positions are modelling choices, not measured dating distributions.

S2. The Murzuq encounter comparison

S2.1 Source and frame

Abuhmida (2013) reports palynology from six Murzuq Basin wells. All 164 appendix samples were transcribed independently, with 158 chart depth and type labels (153 exact matches). The 164 prose-scope rows and 158 chart reports were mapped, retaining barren material and possible caving (`data/units/libya_*.csv`; `docs/audits/2026-10-31-libya-event-map.md`). The primary frame is the chart-coded cuttings design `LY-MZ-CU`. Five wells contribute 59 Llandovery reference cuttings (51 with trilete reports), and three of them contribute 28 older cuttings without trilete reports.

S2.2 Model, targets and qualification

Let $d_i \in \{0, 1\}$ be the reported trilete outcome for eligible cutting i, $c(i)$ its eligibility class, $k(i)$ its well, and $[l_i, h_i]$ its declared age range. Conditional on the fixed stage boundaries, latent age is uniform on that range. Let $r_c > 0$ be the nuisance reference encounter hazard for class c. A mean-one well effect $u_k \sim \text{Gamma}(\nu, \nu)$ is shared by the reference and older cuttings in well k; the primary across-well design fixes $\nu = 2$. Well effects are integrated out, not estimated. Cutting outcomes are independent conditional on their well effect and latent ages.

The first-appearance rule gives $\tau = 443.1$ Ma. Encounter intensity is scaled to $a(t) = 1$ in the five-stage reference window. For a candidate origin $T > \tau$, $a(t) = 0$ for $t > T$. The declared older-to- reference delivery and recovery multiplier is $m_i = \rho$ for transported older cuttings and 1 otherwise, with $\rho \in \{1, 0.5, 0.25, 0.1\}$ fixed in separate analyses rather than estimated. Partition $(\tau, T]$ at stage and numerical-range boundaries. If w_{ip} is the fraction of unit i's age range in piece p, Q_i the fraction in the reference interval at or younger than τ, and x_p the mean of $a(t)$ in piece p, its age-integrated detection probability conditional on u is

$$\pi_i(u; r, x) = Q_i\{1 - \exp(-r_{c(i)} m_i u)\} + \sum_p w_{ip}\{1 - \exp(-r_{c(i)} m_i u x_p)\}.$$

The unshown remainder of the age range lies older than T and contributes zero detection probability. For piecewise-constant histories this expression is exact. The joint likelihood is

$$L(r,x)=\prod_k\int_0^\infty g_{\nu,\nu}(u)\prod_{i:k(i)=k}\pi_i(u;r,x)^{d_i}\{1-\pi_i(u;r,x)\}^{1-d_i}\,du.$$

Thus samples from the same well are marginally dependent. The likelihood profiles r jointly with the history; the reference-only fit supplies starting values, not a plug-in rate held fixed. In the primary data there are 59 reference cuttings (51 positive) and 28 older zeros, across five wells; the older zeros come from three of them. At $T = 459.1$ Ma, 18 older cuttings have all their age support in $(\tau, T]$, while each of the ten Darriwilian cuttings has exposure weight $q_i = 0.9/11.2 = 0.08036$. Under the constant history $a(t) = f$, its zero probability reduces to

$$1-q_i+q_i\exp(-r\rho uf),$$

inside the same well integral as the reference outcomes.

For nonconstant histories the reported target is the fitted-rate-weighted mean intensity

$$\bar a_r(T)=\sum_p c_p(r)x_p,\qquad c_p(r)=\frac{\sum_i r_{c(i)}m_iw_{ip}}{\sum_i r_{c(i)}m_i\sum_q w_{iq}}.$$

The rates in these weights are profiled in the likelihood. With the primary cutting-only design there is one rate class, so class-rate uncertainty does not change the relative piece weights.

Class	Restriction on $a(t)$ for $\tau < t \le T$	Ceiling	Reported use
H0c	$a(t) = x$, constant	1	exact constant-history bound and origin read-out
H0	any path $0 \le a(t) \le 1$	1	bound on $\bar a_r$; pointwise origin and window read-outs
H0m	H0 plus intensity non-decreasing towards τ	1	monotone bound; origin read-out permits the declared 10 Myr delay
H1	any path $0 \le a(t) \le 3$	3	higher-ceiling sensitivity for $\bar a_r$; no origin read-out

For H0c the age-integrated likelihood is exact. For H0, H0m and H1 the implementation bounds each piece using the convex chord for non-detections and Jensen's inequality for detections, then maximises over piece means. With no older detections, as in the primary Murzuq design, the H0 and H1 bounds are attainable. The H0m within-piece relaxation remains conservative because its extremal path need not be monotone inside every piece.

For class K and threshold f, let $\bar\ell_{0,K}(f;T)$ be the supremum of the implemented chord/Jensen **upper log-likelihood envelope** over nuisance rates and piece means satisfying $\bar a_r(T) \ge f$, and let $\ell_{A,K}(T)$ be the exact log likelihood of the best constant history up to that class's ceiling. For H0c the null is also exact, so $\bar\ell_{0,K} = \ell_{0,K}$. For H0 and H1 the envelope is attained in the present no-older-detection application; for H0m it can remain strictly above the likelihood supremum over monotone paths and therefore gives a conservative statistic. The implemented statistic is

$$\Lambda_K(f;T)=\max\left[0,\ 2\{\ell_{A,K}(T)-\bar\ell_{0,K}(f;T)\}\right].$$

The reported $f^*_K(T)$ is the largest f not excluded. H0c, H0m and H0 share the constant alternative on $[0, 1]$; H1 uses its ceiling of 3. The declared critical value is the one-sided count-matched value for the number of older detections plus a simulation-calibrated nonnegative shift. Here there are no older detections and the accepted shift is zero, giving $-2\log(0.05) = 5.991$.

Origin read-outs impose pointwise restrictions rather than only the mean target. H0c requires a constant $a \geq f$ on $(\tau, T]$; H0 requires $f \leq a(t) \leq 1$ throughout that interval; H0m requires the floor from $T - 10$ Ma to τ and permits lower intensity, down to zero, during the first 10 Myr after origin. The separate H0 window read-out beginning at 459.1 Ma requires $a(t) \geq f$ only on $(459.1, T]$ and allows any path in $[0, 1]$ between τ and 459.1 Ma. These are marginal read-outs, not a simultaneous confidence band.

"Qualified" means that the complete screen, first-appearance selection, likelihood fitting and test inversion attained at least the declared 0.93 marginal-coverage floor on the frozen finite simulation grid, with a calibration target of 0.955 and a separate acceptance run. It does not validate uniform age within stages, Gamma$(2, 2)$ well heterogeneity, conditional independence, equal reporting scope, the chosen ρ, or the historical sampling process outside that grid. Inverse-block-size weighting and leave-one-well-out results in S2.5 are retrospective unqualified sensitivities, not alternative qualified dependence models.

The potential screen, eligibility and estimation path were simulated on a declared finite grid. Results are nominal 95% compatibility bounds, qualified per quantity and design against a 0.93 coverage floor (calibration target 0.955), conditional on passing the potential screen, with abstentions counted as covering. The largest observed marginal miss proportion was 38/1,000. At the B4 boundary of the 462.6 Ma window, 20 of 1,000 screen-passing simulations excluded the truth, and 60 delivered non-abstaining window claims, so 20 of those 60 missed. Population 95% coverage, coverage conditional on a non-abstaining result, simultaneous coverage and validity beyond the tested histories are not established. Thirteen Murzuq design/delivery combinations qualify. A fourteenth qualified pair, QB-CO, supplies no older M2 interval and has a narrower class scope. The receipts are in `models/f3/out/qual/`.

S2.3 Contamination and an identification limit

Murzuq's palynofacies fractions use a different preparation and a 200-particle denominator. They cannot supply separate trilete and cryptospore counts for the chart search, so binary observations are retained and contamination enters as a sensitivity model. Let g be the indigenous fraction of reference *encounter intensity* (not of positive samples), and κ the transfer of the contaminant component into older cuttings. For age-exposure weight q_i, an older zero has probability

$$\exp[-ru\kappa(1-g)]\{(1-q_i) + q_i \exp[-ru\rho g f]\}.$$

The likelihood depends on these parameters only through $a = \rho g f$ and $b = \kappa(1-g)$, so tuples preserving both are observationally equivalent within the model. If every $q_i = 1$, only their sum enters the older-zero likelihood. Declining indigenous output can thus be offset by delivery and reference contamination along stated parameter combinations. This is an identification result within the declared model, not a universal property of binary observations.

A predeclared 5,760-row factorial grid varies delivery, indigenous fraction, transfer, allocation, chronology, interval and intensity. Independent uniform ages are compared with fixed, depth-ordered 10%, 50% and 90% positions within source ranges. The threshold $-2\log(0.05)$ is an uncalibrated comparison convention. A contamination scenario that conflicts with older zeros even at $f = 0$ is flagged separately. The 384 break-even entries fix $\kappa = b = 0$. These surfaces and the 36-case well/age influence exercise are unqualified sensitivities.

Table S1. Murzuq sensitivities (`models/f3/loop2/out/`; `docs/loop2-sprint1-results.md`).

Challenge at $\rho = 1$	Observed sensitivity	Scope
Five leave-one-well-out cases	H0 0.1850–0.3646; window endpoints 462.6–468.1 Ma	one factor at a time; unqualified
All nine influence/age/allocation cases	H0 0.1775–0.3646	unqualified
$\kappa = 0$, independent uniform stage ages	critical g: 0.0910 for constant origin intensity; 0.9559, 0.2363, 0.1735 for windows ending 462.6, 467.1, 469.4 Ma	$f = 1$; comparison contours, not confidence limits
Shared midpoint ages	no exposure in the 459.1–462.6 Ma window	fixed chronology scenario
Shared 90% ages	no exposure in the window ending at 467.1 Ma	fixed chronology scenario

S2.4 The calibration material: Borenshult-1

Hooper et al. (2026) base the 459.1 Ma maximum on the oldest possible trilete spores from the Borenshult-1 core, Östergötland, Sweden (Rubinstein and Vajda 2019). Their Supplementary Table 4 keys the soft maximum to *Chelinospora?* sp. at 48.2 m (*Baltoniodus variabilis* zone, lower Sandbian) and takes the value from the GTS2012 Sandbian base plus its uncertainty, 458.4 + 0.7 = 459.1 Ma. The same rule on our timescale gives 458.9 Ma. They did not use the Borenshult spores for a minimum because the trilete marks might be compression artefacts.

Rubinstein and Vajda's Fig. 3 separates definite from possible triletes. It plots definite triletes at six depths from 43.1 to 33.05 m, all Sandbian or younger and the lowest below the 454.41 Ma Kinnekulle K-bentonite, and possible triletes only at 48.2 and 34.0 m. Their Fig. 4 lists no trilete taxon at 41.1 m, so five depths are concordant between the figures. We carry three readings: Fig. 3 definite, Fig. 4 definite and the concordant five, each with the two possible-only depths unresolved, plus Hooper et al.'s wholesale exclusion as a sensitivity (`data/units/borenshult_review/`, `docs/audits/2026-11-05-trilete-evidence-rubric.md`). Symbols and positive depths are not specimen counts.

The readings bear on three separate questions. (1) The maximum: the stage-base rule gives 459.1 Ma from the definite 43.1 m level as from the possible 48.2 m one. (2) A minimum: the definite triletes would set a crown-embryophyte minimum only under an added mapping of their producers to the crown; that mapping is not supplied by the mark, and its minimum would be the young edge of the bentonite age, not the level itself. (3) An absence test: only three Darriwilian samples lie below the first definite trilete. With all three fully exposed, the screen statistic at reference intensity is 1.3157, 1.0753 and 1.1155 for the three readings, against 5.9915, so the core cannot exclude reference-level intensity before its first definite trilete under any reading (`models/f3/out/borenshult_review/screen.csv`). This screen has no numerical chronology and is not a test at 459.1 Ma.

The comparison of the earliest trilete claims on common axes (specimen basis, the describing authors' confidence, independent age control, artefact evidence and documentary consistency) keeps Qusaiba-1 as the secure comparator, retains the Turkish occurrence with its original Ashgillian–Hirnantian age wording, and keeps the Hirnantian Zbrza marks outside strict trilete occurrences because their authors doubt them. No composite score is computed.

S2.5 Time support by well and dependence among cuttings

Fig. S2 plots every chart-coded cutting by well and depth with its full stage enclosure. The older frame has three Darriwilian cuttings in D1-200 and seven in H2-NC186, whose enclosures reach before 459.1 Ma, and 18 Katian or Hirnantian cuttings in D1-200, H2-NC186 and I3-NC186 that are wholly younger. In the encounter likelihood a Darriwilian cutting enters as a mixture: with probability $1-q_i$ its age lies before 459.1 Ma, where a history sustained from 459.1 Ma gives no producers, and with probability q_i it lies inside the window ($q_i = 0.9/11.2$ under a uniform age within the stage).

The qualified model shares a Gamma(2, 2) frailty within each well. Three retrospective, unqualified sensitivities (`models/f3/loop3/murzuq_stage2.py`) ask how much the constant-history contour on ρf depends on cuttings being independent given the shared well effect. Each uses the declared $-2\log(0.05)$ threshold as a comparison convention, not a calibrated critical value.

Treatment	Contour on ρf
Conditionally independent cuttings given shared well frailty (as qualified)	0.0910
A separate reference rate for each well, no frailty	0.0995
Each cutting weighted by 1/(size of its well × formation block)	0.393
Each cutting weighted by 1/(size of its well × stage-support block)	0.419
One well left out at a time	0.093 to 0.186

Table S1b. Murzuq dependence sensitivities (`models/f3/loop3/out/stage2_murzuq/dependence_sensitivity.csv`).

Inverse block-size weights give each block unit total likelihood weight: about seven reference and six older blocks in total weight. Reference blocks mix positive and negative cuttings, so a block is not replaced by a single binary observation. This composite sensitivity does not estimate correlation or full block dependence;

its unqualified comparison contour is about four tenths of reference intensity, against about a tenth for the qualified model. Dropping I3-NC186, which holds ten of the 18 younger older-frame cuttings, has the largest single-well effect. Claude reproduced the four undropped contours with an independent implementation.

S3. QSIM-801: a model-conditional floor calculation

Vecoli et al. (2017, Table 1, p. 321) report 99 cryptospore morphotaxon occurrences across 17 of 20 tabled samples assigned to the late Dapingian–early Darriwilian. Their p. 330 explicitly reports no triletes in the studied sequence. The table was independently reconstructed from the rendered source (`data/units/loop2/replication/`). Its 22 categories include informal morphotypes. Marks are distinct recorded occurrences, not counts of independent producers or dispersal events. The table-defined depths are 675.1–767.8 ft. The methods' shallower endpoint, 586.9 ft, is inconsistent and is not used.

Let θ_i be the trilete/control unit-intensity ratio in sample i, and $h > 0$ a relative effective recovery factor. Suppose the reported objects form a co-ascertained, independent target/control sample. Conditioning Poisson counts on the true land-spore total N_i then gives target probability $h\theta_i/(1 + h\theta_i)$. For a persistent floor $\psi = \min_i \theta_i \geq \theta$, the all-zero probability is at most $(1 + h\theta)^{-\sum_i N_i}$. If each mark contributes a distinct target-detectable unit, $M \leq \sum_i N_i$ permits a conservative rejection rule: no targets and $(1 + h\theta)^{-M} \leq \alpha$. Randomness of M does not invalidate the rule: conditional on the actual totals, rejection is a subset of the all-zero event with enough units to cross the threshold. Then $U = \{\alpha^{-1/M} - 1\}/h$ bounds the *minimum* ratio, not every sample's ratio, a mean, or the pooled share.

Relative effective recovery h	Independent units, $D = 1$	Heuristic $D = 2$	Heuristic $D = 4$
1	0.031	0.062	0.129
0.5	0.061	0.125	0.257
0.25	0.123	0.250	0.515
0.1	0.307	0.624	1.287

Table S2. Model-conditional floor limits; none is an admitted empirical confidence bound. D divides the exponent as a heuristic for clustering, not a validated correction.

The source specifies preparation but not the search coverage needed to show that every mark represents a unit on which triletes were detectable. Independence, effective ascertainment and sampling and stopping conditions are unverified. The calculation therefore stays exploratory. Toy Poisson simulations check the algebra but cannot validate the historical protocol. No matched counted Katian–Llandovery reference was recovered. Qusaiba-1 supplies reference context only. Röstånga and Caspalá/Lipeón are younger than M2, and Hanadir/TAYM-4 is separate material without a qualified design (`manuscript/external_candidates.csv`).

S4. The 515.5 Ma maximum (M1): affinity and the preservation-control argument

Morris et al. (2018b) place the maximum at the base of the *Olenellus* Biozone, about 515.5 Ma, below the oldest possible total-group records (Stage 4 Rome Formation cryptospores), and argue for a real absence before it: land-plant spores share the preservation potential of the marine algal cysts with which they usually occur, so older cyst-bearing sequences without land-plant spores record absence (Morris et al. 2018a). Our earlier drafts described only the placement (`docs/audits/2026-11-07-m1-rationale.md`). The control shows that sporopollenin-walled microfossils survived and were recovered; it does not by itself show comparable terrestrial delivery, coverage of the land areas where early plants grew, sensitivity of preparation and search to small cryptospores, or production of recognisable spores by the earliest producers. The assessment below concerns whether the assembled material can quantify the argument.

No assessed frame admits an E2 older-zero/reference design. The Borenshult 71.1 m zero is secure under the main reading but belongs to a frame that fails its potential screen. It is not a pre-515.5 Ma observation. Strother (2016) places the Tennessee taxa in Charophyta. Taylor and Strother (2024) interpret the packets as pre-embryophytic, and K. Wang, Xu, and Yin (2022) favour ancestral aeroterrestrial algae while calling direct embryophyte affinity unlikely and the interpretation premature. None explicitly accepts the assessed Stage 4 producers as embryophytes. Wall ultrastructure and developmental features bear on precursor relationships, so the inclusive reading stays an explicit alternative. Argentine sample RIOCAP-7999 is unresolved under main and detected under inclusive and outlier readings. Stage 4 converts to 514.5–506.5 Ma, all younger than 515.5 Ma by the maximum's design (`data/units/loop2/m1-m3/`).

S5. The positive body-record analysis

S5.1 Inventory and units

Names. The 241 distinct PBDB macrofossil names overlapping 445.2–393.47 Ma were classified by `gpt-6.1-sol` with web search. Each name received a category, polysporangiophyte and crown-member flags, a basis, a cited source and a confidence (`data/macro/name_classifications.csv`). Names judged implausible for the window, for example the Cretaceous angiosperm leaf genus *Protophyllum*, cannot enter any category. A stratified source audit is in `docs/audits/2026-11-01-loop3-source-review.md`; it does not certify all 241 assignments.

Material identity. Only explicit collection or occurrence identifiers justify replacing a PBDB record with its curated interpretation. Country, taxon and overlapping ages do not establish identity; the pilot used them and falsely merged distinct localities (S7). Regional checklists and unresolved duplicates are withheld from the primary count and kept in a sensitivity. Source decisions are in `data/macro/source_adjudications.csv`. Example: the two Yea horizons are kept as separate units with the full disputed Gorstian–Emsian range, because the Gorstian attribution rests on a disputed graptolite identification. Neither a fixed Gorstian nor a fixed Devonian age is justified.

Units. The primary unit is a locality–age proxy: PBDB formation plus coordinates rounded to 0.1° and identical age support, or a curated named site plus identical age support. For a collapsed unit, the oldest-member age has support from the maximum of member lower ages to the maximum of member upper ages. Missing geographic keys retain the collection identifier. The proxy is conservative but is not a verified bed (`models/f3/loop3/out/primary_unit_membership.csv` lists every constituent record).

Older screen. A separate screen covers all 556 older-only non-spore PBDB rows (69 names). Source review of the seven plausible or unplaced names (17 occurrences) found cyanobacteria, algae or trace fossils and the rejected *Boiophyton*, not an additional embryophyte body. These exclusions are not fossil non-detections, and the screen is not a search denominator (`data/macro/older_records_screen.csv`; `docs/audits/2026-11-01-loop3-older-records-review.md`).

Category admission. Membership is tested by the builder from each record's evidence channel (`data/macro/body_membership.csv`). B1 requires E3 (branched axes bearing sporangia) or nested E5 (crown membership) evidence. Tracheids alone (E4) are synthesis evidence and do not show a branched sporophyte with several sporangia, so E4-only material does not enter B1: the Lochkovian Lang tracheid axis, a sterile axis, stays out. The E3 channel was enumerated only for the Homerian and older, so a younger record without an E3 row was never assessed for branching, and its absence is not a negative. Two such records have published branched fertile material and are admitted to B1 on that evidence: *Cooksonia bohemica* from Kosov, whose emended diagnosis and holotype describe isotomously branched axes with terminal sporangia (Libertín et al. 2022), and the Upper Roman Camp Beds at Capel Horeb, which yielded branching specimens of *Steganotheca striata* and *Cooksonia* sp. from the same Ludlow beds as the tracheid-bearing sterile axis (Edwards and Rogerson 1979, Plate 1e, g). Branching, conducting anatomy and crown membership are adjudicated separately for each, and tracheids are not transferred to the fertile specimens (`docs/audits/2026-11-07-gpt6pro-review2-triage.md`). Separately, PBDB collection 278439 (Xiemisitai Bed 16, Hoboksar County) is in West Junggar, not Tarim (Liu et al. 2024); it is now grouped with the other West Junggar material under Kazakhstania in both the fossil and the effort tables, which changes the region–stage sensitivity (C30) but no locality–age unit.

The Oman sporangial fragments. B0 was declared as any accepted land-plant macro- or mesofossil, sterile axes included (`docs/plan-loop3.md`). The inventory as first built took curated records from the E3–E5 channels only, so the spore-filled sporangial fragments of the Ghaba-1 borehole, Oman (Wellman, Osterloff, and Mohiuddin 2003), recorded in E2 (observation S-w22-K12), never entered it. The primary report describes fragments from three samples of one core, some with a sporangial covering around in situ cryptospores; it dates them Caradoc, and the later review we follow correlates them to the Katian. No associated axes or conducting tissue are preserved, so they stay out of B1 and B2, but the declared B0 admits them. The omission was an inventory-assembly error, so the Oman-inclusive B0 is canonical and the former inventory is reported as an exclusion sensitivity. The fragments enter as one locality–age unit (Katian enclosure 452.8–445.2 Ma), not as twelve fragments or three samples. Codex's membership audit of all 111 E2 observations found no other record that passes the declared B0 before the Silurian (`docs/audits/2026-11-07-b0-membership-review.md`); the older Hawaz sheets lack secure land-plant tissue identity. Their recovery through palynological preparation differs from the macrofossil collections that dominate the recording proxies, which is a further reason the B0 histories may misdescribe the sequence. The B0 values now come from the canonical `build_reviewed.py` and `analysis.py` pipeline. The explicit `body_membership.csv` admits K12 as B0 only and excludes the Hawaz candidate on tissue identity; `variant=without_oman_fragments` reconstructs the former inventory.

Bounds and allocation diagnostics for both inventories come from the same current analysis. The earlier additive outputs are retained as archived provenance, with their receipt bound to historical commit `df805b2`.

Reading	Oman	Units	Oldest	Const.	PBDB	PBDB −plant	Decl. 25	Decl. 10	−plant × decl. 10
main	included	76/78	452.8	455.19	456.62	456.09	465.21	none	none
main	excluded	75/77	432.9	434.51	436.38	435.73	437.16	none	none
inclusive	included	80/82	452.8	455.06	456.43	455.92	464.40	none	none
inclusive	excluded	79/81	445.2	447.20	448.27	447.85	452.98	none	none

Table S3b. Land-plant body (B0) bounds (Ma) with the Katian Oman fragments included (canonical) and excluded, Emsian cutoff. Units: guaranteed/possible; Oldest: oldest possible age. History abbreviations: Const., constant; PBDB, proportional to PBDB land and marginal-marine collection density; PBDB −plant, the same with plant-matched collections removed; Decl. 25 and Decl. 10, exponential decline into the past on 25 and 10 Myr timescales; −plant × decl. 10, their product. "None": no bound within the 469.4 Ma limit of the declared domain.

With the fragments, the oldest unit is Katian, followed by more than 12 Myr with no body unit, and every declared history fits the sequence poorly (Table S7). The bounds move about 20 Myr older. Under the inclusive reading without Oman, the Hirnantian Zbrza axes are the oldest unit and give 447.20 Ma under constant intensity. Neither inventory supplies a lag or delay: the 458.2 Ma cryptospore value is a fossil minimum, not an origin, and the bounds are model-conditioned endpoints. The comparison shows how far an endpoint bound depends on its oldest admitted unit.

S5.2 The endpoint calculation

For a fixed younger cutoff c, let $U(a) = \int_c^a w(s)\,ds$ be cumulative recovery. Conditional on n independent units with ages drawn with density proportional to w on $[c, T]$, the maximum A satisfies $P(A \le a) = \{U(a)/U(T)\}^n$. Inverting at 0.05 gives the exact one-sided 95% older bound

$$U(T_{\text{upper}}) = U(A)\,0.05^{-1/n}.$$

The absolute recovery rate cancels. For constant w this is the Strauss and Sadler (1989) and Marshall (1990) expression with a fixed cutoff, which uses all n observations. Marshall (1994, 1997) and S. C. Wang et al. (2016) relax the assumption of uniform recovery in other ways. A bound beyond the declared history's domain (469.4 Ma, where the held PBDB effort series ends) is reported as *not bounded within the domain*, not as an infinite-age inference.

Age envelope. The primary age treatment uses the greatest possible oldest age and the smallest number of units guaranteed older than c. The bound increases with A and decreases with n, so this is an envelope of the exact-age bounds whenever every reported age interval contains the true age. That geological condition is separate from statistical coverage. Independent-uniform and shared-quantile allocations (256 draws, case-keyed seeds rooted at 20261102) are reported as sensitivities. Their quantiles are not confidence bounds (Table S8).

Histories. Six histories are fixed without fitting them to the focal fossils:

1. constant recovery in calendar time;
2. PBDB all-taxa land and marginal-marine collections per Myr, apportioned by age overlap;
3. the same, with collections matching the held plant-filtered extract removed;
4. exponential decline into the past, 25 Myr timescale;
5. exponential decline into the past, 10 Myr timescale;
6. the plant-excluded density multiplied by the 10 Myr decline.

Plant exclusion is a response-related filter, not verified independent effort. No proxy establishes actual search opportunities. Cutoffs are the ends of the Emsian, Pragian, Lochkovian and Pridoli.

Shape diagnostics. Conditional on the maximum, the remaining unit ages should be distributed as $U(a)/U(A)$. Kolmogorov–Smirnov statistics at midpoint ages and across age draws expose histories that fit the observed sequence poorly (Table S7). Table S7 reports every declared age treatment: midpoint ages, independent uniform draws within each range and a shared quantile applied to all ranges. No probability law for ages within ranges was declared, so none is preferred. For crown vascular-plant bodies, constant

recovery fits under independent draws but not at midpoint ages or shared quantiles; the 10 Myr decline fits poorly under all three. The robust result is discrimination against steep declines, not support for constant recovery. The p values are asymptotic Kolmogorov–Smirnov values computed on allocated ages treated as exact, not exact finite-sample tests, and their spread across draws is a set of percentiles, not a combined test for interval-dated fossils, a model-selection criterion or an empirical calibration. **Interval feasibility.** A history that fails under the allocations drawn may fit another allocation that the ranges admit; the GPT-6 Pro review constructed such allocations for the 25 Myr decline. For each category and history, `models/f3/loop3/interval_feasibility.py` finds the admissible allocation that minimises the Kolmogorov–Smirnov distance D. Each unit's age support becomes an interval of transformed exposure; compatibility of a unit with an order-statistic rank changes only at finitely many values of the maximum's exposure, so the search enumerates every such event point and the open cells between them and solves a bipartite matching in each, bracketing the global minimum of D to within 5×10^{-11}. The two units in each category whose support straddles the Emsian cutoff may fall below it and leave the conditional sample, so all four inclusion states are searched. The cutoff is held at 393.47 Ma; shared boundary shifts are treated separately (S5.3). Table S7b displays, for each state and history, the finite-sample p at the best allocation, and flags where the asymptotic reference would change the classification; both values are in the source CSV. Where no allocation reaches 0.05 these are upper bounds, so the failure holds for every allocation within the recorded supports; where one does, the allocation file is a witness.

What a failure and a pass mean. Fix the recording history, cutoff, timescale, inventory and unit rule. Let z be the latent exact ages and inclusion state, S the allocations allowed by the recorded supports, and $p(z)$ the finite-sample conditional-on-count-and-maximum KS p value under independent continuous occurrence ages. The interval rule rejects only if every $z' \in S$ has $p(z') < \alpha$. Under an observation model in which the reported intervals contain the latent ages almost surely, the true allocation belongs to S, so this rejection event is a subset of rejection by the level-α exact-age test. It is therefore conservative. If containment can fail with probability δ, the direct size bound is instead $\alpha + \delta$; these geological ranges have no demonstrated joint coverage. The maximisation resembles maximised-p constructions (Berger and Boos 1994), but here the intervals coarsen observations rather than form a confidence set for a nuisance parameter. Enumerating all cutoff-inclusion states preserves the argument when the true state is included. A passing witness is an admissible allocation, not an estimated chronology, evidence of model adequacy or a ranking of histories. A coherent section chronology could remove a passing allocation but cannot rescue an interval-wide failure if it contains the truth. A separate brute-force permutation implementation (Codex) agreed on 720 fixed-D decisions and 120 optimum brackets, and Claude recomputed D from the decisive witness files; the GPT-6 Pro follow-up review independently checked small cases and all 48 saved witnesses.

Category	Units	Const.	PBDB	PBDB −plant	Decl. 25	Decl. 10	−plant × decl. 10
B2	48 guaranteed	0.985	0.722	0.256	0.035*	3e-06*	0.045†
B2	49 (one straddling)	0.994	0.667	0.217	0.049†	6e-06*	0.051
B2	50 possible	0.998	0.612	0.184	0.067	9e-06*	0.057
B0 +Oman	76 guaranteed	3e-07*	9e-05*	1e-06*	0.002*	5e-17*	6e-08*
B0 +Oman	77 (one straddling)	2e-07*	8e-05*	8e-07*	0.003*	1e-16*	9e-08*
B0 +Oman	78 possible	2e-07*	7e-05*	6e-07*	0.004*	3e-16*	1e-07*

Table S7b. Best attainable conditional-age diagnostic over all admissible allocations, main reading, Emsian cutoff: finite-sample p at the most favourable allocation. * no admissible allocation reaches $p = 0.05$, so the history fails this diagnostic for every age assignment within the recorded supports; † fails under the finite-sample reference but reaches 0.05 under the asymptotic one. Units: number inside the window, by inclusion of the units that straddle the cutoff (either of the two gives the same result). A failure is a conservative rejection only if the supports contain the true ages (see text); a pass is a possibility, not a fit (`models/f3/loop3/out/stage2_interval/`).

Under the finite-sample reference, for crown-member bodies the 10 Myr decline fails in every inclusion state. The 25 Myr decline passes with all 50 possible units, falls just short with one straddling unit ($p = 0.049$, which passes only asymptotically) and fails with the 48 guaranteed units. The plant-excluded density times the 10 Myr decline passes with 49 or 50 units but not 48. For canonical land-plant bodies, every history fails in every state under either reference.

S5.3 Results and sensitivities

Table labels. B0 +Oman is the canonical land-plant body category; B0 -Oman excludes the Ghaba-1 fragments; B1 is the E3-or-E5 polysporangiophyte category; B2 is crown vascular-plant membership. Const. is constant intensity; PBDB is all-collection density; PBDB -plant removes plant-matched collections; Decl. 25/10 denotes the exponential time constant in Myr. Indep. is independent-uniform age allocation; shared q places every unit at a common within-range quantile.

Reading	Category	Units	Oldest	Const.	PBDB	PBDB −plant	Decl. 25	Decl. 10	−plant × decl. 10
main	B0 +Oman	76/78	452.8	455.19	456.62	456.09	465.21	none	none
main	B1	69/71	432.9	434.65	436.68	435.98	437.57	none	none
main	B2	48/50	426.7	428.84	428.88	428.45	431.63	none	none
inclusive	B0 +Oman	80/82	452.8	455.06	456.43	455.92	464.40	none	none
inclusive	B1	73/75	445.2	447.37	448.53	448.07	453.76	none	none
inclusive	B2	53/55	426.7	428.63	428.67	428.28	431.10	none	468.65

Table S3. Conservative 95% older bounds (Ma) by category, reading and recording history from the canonical inventory, in which B0 includes the Oman fragments (Table S3b compares B0 with and without them); Emsian cutoff. Units: guaranteed/possible; Oldest: oldest possible age. History abbreviations: Const., constant; PBDB, proportional to PBDB land and marginal-marine collection density; PBDB −plant, the same with plant-matched collections removed; Decl. 25 and Decl. 10, exponential decline into the past on 25 and 10 Myr timescales; −plant × decl. 10, their product. "None": no bound within the 469.4 Ma limit of the declared domain.

Cutoff (Ma)	Units	Const.	PBDB	PBDB −plant	Decl. 25	Decl. 10	−plant × decl. 10
Emsian (393.47)	48	428.84	428.88	428.45	431.63	none	none
Pragian (410.62)	25	428.75	429.71	429.43	429.75	433.80	451.56
Lochkovian (413.02)	18	429.18	430.09	429.81	430.24	434.25	447.52
Pridoli (419.62)	13	428.53	428.54	428.35	428.92	429.80	429.55

Table S4. Crown vascular-plant bounds (Ma) by younger cutoff, main reading. Abbreviations as in Table S3.

B2 unit or source variant	Units	Constant	PBDB density	Decline 25 Myr
Individual collections/observations	78	428.00	428.03	429.58
Locality–age proxies (primary)	48	428.84	428.88	431.63
Whole localities	47	428.89	428.93	431.75
Region–stage cells	19	432.38	432.16	442.78
Wide-age records restored	73	428.09	428.12	429.79
Uncertain material restored	54	428.60	428.63	431.01
Low-confidence names omitted	46	428.94	428.98	431.87
Weak botanical source omitted	48	428.84	428.88	431.63
Kosov/Podolia links collapsed	47	428.89	428.93	431.75
Curated oldest-record set only	15	434.05	434.62	450.50

Table S5. Crown vascular-plant bounds under alternative units and source choices (main reading, Emsian cutoff). Region–stage cells use midpoint stage and are a deliberately coarse dependence diagnostic.

Region left out	Units	Oldest possible	Constant	PBDB density
Gondwana Australia Antarctica	44	425.0	427.22	427.33
Gondwana South America	47	426.7	428.89	428.87
peri Gondwana Europe	47	426.7	428.89	428.85

Region left out	Units	Oldest possible	Constant	PBDB density
Baltica	45	426.7	428.99	431.80
Avalonia	44	426.7	429.04	428.95
Kazakhstania	44	426.7	429.04	429.08
South China	43	426.7	429.10	429.06
multiple	42	426.7	429.16	428.74
Laurentia	28	426.7	430.45	429.58

Table S6. Leave-one-region-out bounds for crown vascular-plant members, with PBDB exposure recomputed after the same omission. Regions are approximate modern country/state assignments.

Shared timescale boundaries. The bounds above take stage boundaries as fixed. Five deterministic scenarios (`models/f3/loop3/timescale_sensitivity.py`) move every boundary in `data/timescale.csv` at once: baseline; all one quoted uncertainty younger; all one older; and two alternating patterns, scaled to 0.7345 and 0.5610 of the quoted uncertainties so that stage order is kept. Each shifted boundary moves curated stage-based endpoints, the cutoff, the analysis domain and the effort-stage widths together. A PBDB numerical interval moves only when both of its endpoints match project boundaries (27 of 35 PBDB crown-member units); otherwise its numbers are kept, because an equal number does not show that the source used the ICS boundary. The maximum is compared three ways: the published 431.2 Ma held fixed; 431.2 Ma moved to the same fractional position between the shifted Homerian and Sheinwoodian bases; and Hooper et al.'s rule reapplied on the current scale, the shifted Homerian base plus the 1.3 Myr uncertainty the current table quotes for it (431.9 Ma at baseline, against the 431.2 Ma their GTS2012 value gives). The constant-intensity bound stays younger than the maximum under all three comparisons in every scenario, with margins of −4.65 to −0.71 Myr; the two collection-density histories also stay younger. The 25 Myr decline crosses in both directions. Scenarios with an older cutoff have 47 guaranteed units because one PBDB support (393.47–399.5 Ma) keeps its numerical younger end. These are sensitivity scenarios, not a probability distribution over geological ages.

Scenario	Cutoff	Units	Const.	PBDB	PBDB −plant	Decl. 25	431.2 mapped	Source rule
baseline	393.47	48	428.84	428.88	428.45	431.63	431.20	431.90
all younger	392.48	48	427.31	427.53	427.08	429.98	429.93	430.60
all older	394.46	47	430.42	430.35	429.90	433.41	432.47	433.20
alternating a	394.20	47	430.01	429.05	428.78	432.96	430.72	430.95
alternating b	392.91	48	427.98	428.64	428.11	430.70	431.56	432.63

Table S6b. Crown vascular-plant bounds (Ma) under shared boundary scenarios, main reading, Emsian cutoff (shifted with the scenario). Units: guaranteed. "431.2 mapped": the published maximum held at its fractional position in the shifted Homerian; "Source rule": shifted Homerian base plus its quoted 1.3 Myr uncertainty (`models/f3/loop3/out/stage2_timescale/`).

Influential records. Leave-group-out comparisons at the same unit rule and cutoff (`models/f3/loop3/out/influential_`
remove the records that set the oldest ages. Under constant intensity, removing the three Yea *Baragwanathia* units changes the crown-member bound by −1.67 Myr (younger): the oldest age falls to the Bathurst Island 425.0 Ma support. Removing the five Bathurst Island units changes it by 0.26 Myr (older), because Yea remains oldest while the count falls. Removing *C. barrandei* changes the polysporangiophyte bound by −2.38 Myr but canonical B0 by only 0.03 Myr, because the Oman unit sets B0's oldest age. Under other histories and readings the signs can change, since loss of count and loss of the oldest record pull in opposite directions. A leave-out is a leverage diagnostic, not a reidentification of any fossil.

Category	History	Nominal p, midpoint ages	Independent-uniform ages, 5–95%	Shared-quantile ages, 5–95%
B0 +Oman	Const.	5.5e-12	3.1e-13–4e-09	2.9e-12–1.2e-11
B0 +Oman	PBDB	7.5e-08	9.1e-09–4.1e-06	2.1e-09–1.6e-07
B0 +Oman	PBDB −plant	1.1e-10	9.4e-12–1.6e-08	2.5e-12–2.7e-10

Category	History	Nominal p, midpoint ages	Independent-uniform ages, 5–95%	Shared-quantile ages, 5–95%
B0 +Oman	Decl. 25	6.3e-06	3.8e-06–0.00086	2e-16–0.00045
B0 +Oman	Decl. 10	7.4e-20	1.2e-21–8.5e-17	1e-39–2.8e-16
B0 +Oman	−plant × decl. 10	1.2e-08	1.1e-09–1.3e-07	4.6e-16–6.3e-08
B0 -Oman	Const.	0.0066	0.004–0.055	1.1e-09–0.0062
B0 -Oman	PBDB	0.02	0.0095–0.091	0.00033–0.022
B0 -Oman	PBDB −plant	0.0019	0.00074–0.03	4.4e-05–0.0047
B0 -Oman	Decl. 25	1.2e-07	4.9e-09–3.2e-06	2e-21–1.9e-06
B0 -Oman	Decl. 10	2.5e-20	4.3e-22–3.7e-17	1.4e-40–8.4e-17
B0 -Oman	−plant × decl. 10	3.9e-09	3.2e-10–5.2e-08	1.7e-16–1.4e-08
B1	Const.	0.006	0.0058–0.082	8.7e-09–0.0063
B1	PBDB	0.017	0.011–0.14	0.00049–0.034
B1	PBDB −plant	0.0014	0.00095–0.032	4.6e-05–0.0043
B1	Decl. 25	5.9e-07	5.6e-08–2.1e-05	1.6e-19–2e-05
B1	Decl. 10	1.6e-18	2.9e-20–5.1e-15	4.4e-37–1.4e-14
B1	−plant × decl. 10	1.7e-07	7.5e-09–1.2e-06	5.2e-15–4e-07
B2	Const.	0.0085	0.056–0.71	3.1e-09–0.016
B2	PBDB	0.0092	0.016–0.24	2.7e-05–0.058
B2	PBDB −plant	8.6e-05	0.00022–0.019	2.2e-06–0.019
B2	Decl. 25	4.9e-06	4.9e-05–0.016	4.6e-16–0.0019
B2	Decl. 10	3.1e-13	1.6e-11–5.1e-07	3.5e-26–8e-07
B2	−plant × decl. 10	1.5e-05	0.001–0.042	5.9e-11–0.0064

Table S7. Within-record shape diagnostics (conditional-on-maximum Kolmogorov–Smirnov), main reading, under each declared age treatment. Ranges are the 5th–95th percentiles of nominal p across 256 age draws, not minimum-to-maximum ranges.

Category	History	Allocation	5%	Median	95%	Unbounded fraction
B0 +Oman	Const.	indep.	447.70	451.38	454.82	0.00
B0 +Oman	PBDB	indep.	448.82	452.47	456.20	0.00
B0 +Oman	Decl. 25	indep.	453.81	459.13	464.39	0.00
B0 +Oman	Const.	shared q	447.67	451.19	454.74	0.00
B0 +Oman	PBDB	shared q	448.78	452.27	456.10	0.00
B0 +Oman	Decl. 25	shared q	453.89	458.73	464.17	0.00
B0 -Oman	Const.	indep.	432.27	433.50	434.38	0.00
B0 -Oman	PBDB	indep.	432.04	433.84	436.06	0.00
B0 -Oman	Decl. 25	indep.	434.48	435.91	436.97	0.00
B0 -Oman	Const.	shared q	432.23	433.27	434.35	0.00
B0 -Oman	PBDB	shared q	432.00	433.27	435.99	0.00
B0 -Oman	Decl. 25	shared q	434.48	435.66	436.90	0.00
B1	Const.	indep.	432.35	433.32	434.51	0.00
B1	PBDB	indep.	432.10	433.31	436.33	0.00
B1	Decl. 25	indep.	434.79	435.93	437.35	0.00
B1	Const.	shared q	432.36	433.40	434.48	0.00
B1	PBDB	shared q	432.11	433.55	436.28	0.00
B1	Decl. 25	shared q	434.85	436.03	437.28	0.00
B2	Const.	indep.	424.24	426.24	427.69	0.00

Category	History	Allocation	5%	Median	95%	Unbounded fraction
B2	PBDB	indep.	425.05	426.64	427.80	0.00
B2	Decl. 25	indep.	426.13	428.43	430.19	0.00
B2	Const.	shared q	421.58	424.10	427.03	0.00
B2	PBDB	shared q	421.22	424.81	427.18	0.00
B2	Decl. 25	shared q	423.10	425.92	429.34	0.00

Table S8. Age-allocation sensitivity (Emsian cutoff, main reading). Quantiles across allocations are descriptive, not confidence bounds.

S5.4 Recovery before the oldest record

After seeing that steep smooth declines fit poorly, we evaluated a complete stage-break grid: flat and PBDB baselines; breaks at 419.62, 426.7 and 432.9 Ma; older multipliers 1, 0.5, 0.1 and 0.01; all categories and both readings, with an additional B0 exclusion inventory (192 cells). The 426.7 Ma break coincides with the oldest possible main B2 record, so no main B2 unit's age support is changed and the within-record shape statistic is unchanged. The grid is a retrospective sensitivity; the 0.05 rule supplies no nominal, simultaneous or selection-adjusted confidence for it. The 432.9 Ma break is a null control when the bound lies younger. No member was selected to match a clock (Table S9).

Baseline	Break (Ma)	Older multiplier	Changes age support?	Older bound (Ma)	Midpoint shape p
constant	419.62	1	True	428.84	0.00848
constant	419.62	0.5	True	430.52	0.00424
constant	419.62	0.1	True	444.00	0.00223
constant	419.62	0.01	True	none	0.00191
constant	426.7	1	False	428.84	0.00848
constant	426.7	0.5	False	430.98	0.00848
constant	426.7	0.1	False	448.10	0.00848
constant	426.7	0.01	False	none	0.00848
constant	432.9	1	False	428.84	0.00848
constant	432.9	0.5	False	428.84	0.00848
constant	432.9	0.1	False	428.84	0.00848
constant	432.9	0.01	False	428.84	0.00848
PBDB density	419.62	1	True	428.88	0.00918
PBDB density	419.62	0.5	True	430.60	0.0186
PBDB density	419.62	0.1	True	451.09	0.00729
PBDB density	419.62	0.01	True	none	0.00576
PBDB density	426.7	1	False	428.88	0.00918
PBDB density	426.7	0.5	False	430.98	0.00918
PBDB density	426.7	0.1	False	458.47	0.00918
PBDB density	426.7	0.01	False	none	0.00918
PBDB density	432.9	1	False	428.88	0.00918
PBDB density	432.9	0.5	False	428.88	0.00918
PBDB density	432.9	0.1	False	428.88	0.00918
PBDB density	432.9	0.01	False	428.88	0.00918

Table S9. Retrospective stage-break grid for crown vascular-plant members, main reading (`models/f3/loop3/out/suppres`

The age–intensity frontier. The grid fixes multipliers and reads off bounds. The frontier inverts it: for a candidate lineage age T and an assumed delay d before recoverable diagnostic bodies, what ratio of earlier to later recording intensity puts the corresponding recoverability start $T - d$ exactly on the 95% bound? With constant later intensity, a step at the oldest possible age A, fixed cutoff c and n guaranteed units, the earlier exposure needed is $(A - c)(0.05^{-1/n} - 1)$, and the ratio is that exposure divided by $T - d - A$ (`models/f3/loop3/frontier.py`). A ratio above one means no reduction is needed. Where $T - d \leq A$ the contour is undefined in this parameterisation (shown as —), which does not exclude the combination, because the oldest fossil may be younger within its support. The step's position is chosen from the data and the contour is not a fitted rate, a likelihood surface or a joint confidence region. The delay and the earlier rate are not separately identifiable from positive records: a longer delay needs less suppression for the same T.

Units	Delay (Myr)	431.2 Ma	440 Ma	444 Ma	447 Ma	452 Ma
Locality–age (primary)	0	0.476	0.161	0.124	0.105	0.085
Locality–age (primary)	5	—	0.258	0.174	0.140	0.105
Locality–age (primary)	10	—	0.648	0.293	0.208	0.140
Locality–age (primary)	20	—	—	—	7.133	0.404
Region–stage	0	1.261	0.427	0.328	0.280	0.224
Region–stage	5	—	0.684	0.461	0.371	0.280
Region–stage	10	—	1.720	0.777	0.551	0.371
Region–stage	20	—	—	—	18.917	1.071
Curated oldest set	0	1.632	0.552	0.425	0.362	0.290
Curated oldest set	5	—	0.885	0.597	0.480	0.362
Curated oldest set	10	—	2.226	1.006	0.713	0.480
Curated oldest set	20	—	—	—	24.486	1.386

Table S9b. Retrospective age–intensity frontier for crown vascular-plant members, main reading, Emsian cutoff: earlier/later recording intensity at which each candidate lineage age lies on the conservative 95% bound, by unit treatment and assumed delay (`models/f3/loop3/out/frontier.csv`). Primary units: 48 guaranteed; region–stage: 19; curated: 15; oldest possible age 426.7 Ma in each.

S5.5 Biological delay

Fixed delays of 0, 5, 10 and 20 Myr map the recoverability bound to a lineage bound and shift the lineage domain by the same amount. A delay is an assumed mapping, not inferred from this record or transferred from the spore comparison (Table S10).

Category	Assumed delay (Myr)	Recoverability bound	Lineage bound
B0 +Oman	0	455.19	455.19
B0 +Oman	5	455.19	460.19
B0 +Oman	10	455.19	465.19
B0 +Oman	20	455.19	475.19
B1	0	434.65	434.65
B1	5	434.65	439.65
B1	10	434.65	444.65
B1	20	434.65	454.65
B2	0	428.84	428.84
B2	5	428.84	433.84
B2	10	428.84	438.84
B2	20	428.84	448.84

Table S10. Fixed-delay mapping, constant recovery, main reading.

S5.6 Simulation checks

Each of 144 cells has 3,000 replicates (seed 20261101) at the reviewed unit counts and recoverability endpoints of 432, 445 and 458.2 Ma. Each correctly specified history is checked with exact and stage-censored dates. The checks then mismatch constant and steeply declining histories, repeat underlying units two or four times, and

impose 10 or 20 Myr delays with and without the correct adjustment. Coverage counts no-bound outcomes as covering. Claim frequency and coverage among bounded claims are reported separately, because 95% unconditional coverage does not imply 95% coverage after selecting bounded outcomes. These runs verify the declared procedures and expose their failure modes. They do not qualify the fossils (Table S11; all categories in `simulation_results.csv`).

Scenario	True	Fitted	Ages	Dup.	True delay	Adj.	Claims	Coverage (95% CI)
matched	Const.	Const.	exact	1	0	0.0	1.000	0.955 (0.947–0.962)
matched	Const.	Const.	stage envelope	1	0	0.0	1.000	1.000 (0.998–1.000)
matched	Const.	Const.	exact	1	0	0.0	1.000	0.949 (0.941–0.957)
matched	Const.	Const.	stage envelope	1	0	0.0	1.000	0.985 (0.980–0.989)
matched	Const.	Const.	exact	1	0	0.0	1.000	0.945 (0.936–0.953)
matched	Const.	Const.	stage envelope	1	0	0.0	1.000	0.983 (0.977–0.987)
matched	Decl. 25	Decl. 25	exact	1	0	0.0	1.000	0.945 (0.936–0.952)
matched	Decl. 25	Decl. 25	stage envelope	1	0	0.0	1.000	0.956 (0.948–0.963)
matched	Decl. 25	Decl. 25	exact	1	0	0.0	1.000	0.958 (0.951–0.965)
matched	Decl. 25	Decl. 25	stage envelope	1	0	0.0	1.000	0.990 (0.985–0.993)
matched	Decl. 25	Decl. 25	exact	1	0	0.0	0.212	0.941 (0.932–0.949)
matched	Decl. 25	Decl. 25	stage envelope	1	0	0.0	0.070	0.966 (0.959–0.972)
matched	Decl. 10	Decl. 10	exact	1	0	0.0	0.139	0.950 (0.942–0.958)
matched	Decl. 10	Decl. 10	stage envelope	1	0	0.0	0.078	0.998 (0.996–0.999)
matched	Decl. 10	Decl. 10	exact	1	0	0.0	0.065	0.949 (0.940–0.956)
matched	Decl. 10	Decl. 10	stage envelope	1	0	0.0	0.034	0.966 (0.959–0.972)
matched	Decl. 10	Decl. 10	exact	1	0	0.0	0.053	0.950 (0.941–0.957)
matched	Decl. 10	Decl. 10	stage envelope	1	0	0.0	0.032	0.968 (0.961–0.974)
matched	PBDB	PBDB	exact	1	0	0.0	1.000	0.950 (0.942–0.958)
matched	PBDB	PBDB	stage envelope	1	0	0.0	1.000	0.999 (0.998–1.000)
matched	PBDB	PBDB	exact	1	0	0.0	1.000	0.957 (0.949–0.963)
matched	PBDB	PBDB	stage envelope	1	0	0.0	1.000	0.967 (0.960–0.973)
matched	PBDB	PBDB	exact	1	0	0.0	1.000	0.953 (0.945–0.960)
matched	PBDB	PBDB	stage envelope	1	0	0.0	1.000	0.999 (0.998–1.000)
matched	PBDB −plant	PBDB −plant	exact	1	0	0.0	1.000	0.954 (0.946–0.961)
matched	PBDB −plant	PBDB −plant	stage envelope	1	0	0.0	1.000	1.000 (0.998–1.000)

Scenario	True	Fitted	Ages	Dup.	True delay	Adj.	Claims	Coverage (95% CI)
matched	PBDB −plant	PBDB −plant	exact	1	0	0.0	1.000	0.948 (0.939–0.955)
matched	PBDB −plant	PBDB −plant	stage envelope	1	0	0.0	1.000	0.983 (0.977–0.987)
matched	PBDB −plant	PBDB −plant	exact	1	0	0.0	1.000	0.952 (0.944–0.959)
matched	PBDB −plant	PBDB −plant	stage envelope	1	0	0.0	1.000	0.954 (0.946–0.961)
matched	−plant × decl. 10	−plant × decl. 10	exact	1	0	0.0	0.175	0.951 (0.943–0.958)
matched	−plant × decl. 10	−plant × decl. 10	stage envelope	1	0	0.0	0.119	0.992 (0.988–0.995)
matched	−plant × decl. 10	−plant × decl. 10	exact	1	0	0.0	0.069	0.954 (0.946–0.961)
matched	−plant × decl. 10	−plant × decl. 10	stage envelope	1	0	0.0	0.049	0.951 (0.942–0.958)
matched	−plant × decl. 10	−plant × decl. 10	exact	1	0	0.0	0.045	0.956 (0.948–0.962)
matched	−plant × decl. 10	−plant × decl. 10	stage envelope	1	0	0.0	0.040	0.960 (0.953–0.967)
history mismatch	Const.	Decl. 10	exact	1	0	0.0	0.000	1.000 (0.999–1.000)
history mismatch	Const.	Decl. 10	exact	1	0	0.0	0.000	1.000 (0.999–1.000)
history mismatch	Const.	Decl. 10	exact	1	0	0.0	0.000	1.000 (0.999–1.000)
history mismatch	Decl. 10	Const.	exact	1	0	0.0	1.000	0.241 (0.226–0.257)
history mismatch	Decl. 10	Const.	exact	1	0	0.0	1.000	0.104 (0.094–0.116)
history mismatch	Decl. 10	Const.	exact	1	0	0.0	1.000	0.034 (0.028–0.041)
duplicated horizons	Const.	Const.	exact	2	0	0.0	1.000	0.782 (0.767–0.797)
duplicated horizons	Const.	Const.	exact	4	0	0.0	1.000	0.529 (0.511–0.546)
recoverability delay unadjusted	Const.	Const.	exact	1	10	0.0	1.000	0.000 (0.000–0.001)
recoverability delay adjusted	Const.	Const.	exact	1	10	10.0	1.000	0.947 (0.938–0.954)
recoverability delay unadjusted	Const.	Const.	exact	1	20	0.0	1.000	0.000 (0.000–0.001)
recoverability delay adjusted	Const.	Const.	exact	1	20	20.0	1.000	0.952 (0.943–0.959)

Table S11. Simulation results for crown vascular-plant counts. True and Fitted: generating and fitted recording histories (abbreviations as in Table S3); Dup.: duplication factor; True delay: imposed delay (Myr) between lineage origin and the start of recording; Adj.: delay adjustment applied in the fit; Claims: fraction of replicates giving a bound; Coverage: of the lineage or recording endpoint as targeted, with Wilson 95% interval.

S5.7 Figures from the reviewed analysis

Figs S4–S6 are the reviewed analysis's own diagnostic figures (`models/f3/loop3/figures_reviewed.py`), regenerated from the accepted tables: recovery bounds by category, the crown vascular-plant sensitivity grid and the retrospective suppression grid.

S6. From specimens to nodes

The specimen-to-node pilot separates specimen observations, accepted event readings, encounter, trait persistence and node interpretation (`data/pilot/`, `docs/pilot-traits.md`). We reused its source-adjudicated combinations, Hooper et al.'s deposited crown-state outputs and the checked stochastic-map analyses without rerunning them. Aperture (X21) and dispersal unit (X26) were modelled separately, so their marginal supports do not define a joint ancestral phenotype. The X26 monad category combines trilete and inaperturate monads; it is not an automatic absence of cryptospory.

For M2, a proximal trilete aperture is modal at the embryophyte crown across the deposited methods and trees. X26 has a dyad-dominated alternative while X21 stays trilete-dominated, so no joint phenotype probability follows. On the converged extant-tree runs, 77 of 3,382 trilete starts depart along the path to crown Tracheophyta: conditional persistence 0.977 (Monte Carlo SE 0.0026). These trees put crown Embryophyta at 508 Ma, not at Hooper et al.'s 457–450 Ma. Fixed-rate retention scenarios to 452.8 Ma give an extant-tree median of 0.989 (95% interval 0.895–0.995) from 459.1 Ma and 0.894 (0.332–0.956) from 515.5 Ma. These are model-conditional aperture retention scenarios, not observed survival, and they are not multiplied into any coverage.

For M3, the *Cooksonia barrandei* holotype links trilete production and multiple sporangia directly, but its tracheids are unknown. The Irish Homerian material has a conditional sporangial interpretation and no direct demonstration of the anatomy known from younger *C. pertoni*. A total-group or stem occurrence and a crown assignment remain distinct. For M1, cryptospore persistence has not been assessed.

S7. Analyses superseded during review

Analyses and wording were replaced after independent review. The record is kept so that readers can see what changed and why.

The Loop 3 body-record pilot (`docs/loop3-pilot.md`, provisional, by Claude) reported logistic-ramp limits of 452.6, 454.1 and 438.0 Ma for B0, B1 and B2. It read them as favouring the young clock, with a 15–25 Myr preservation lag. Codex's review found:

- stage collection totals passed where per-Myr density was required, counting stage duration twice. The unit correction alone moved B0's logistic limit from 452.57 to 436.78 Ma and flipped the selected capacity between grid boundaries;
- medians of imputed-age limits presented as 95% limits;
- false replacement of distinct PBDB localities through country, genus and age overlap;
- collection numbers treated as independent beds;
- a category gate that let names judged implausible into B1 and B2;
- two different papers sharing the source identifier `wang2026`.

All were repaired. The fixed-history endpoint calculation (S5.2) replaced the profiled ramp. Its numbers and interpretation govern; the pilot's do not (`docs/audits/2026-11-01-loop3-methods-review.md`, `docs/loop3-results.md`).

The GPT-6 Pro review (`docs/audits/2026-11-07-gpt6pro-triage.md`). An independent review of commit `60c406b` found that the earlier drafts described the 515.5 Ma maximum as resting only on affinity, omitting the published preservation-control argument; kept the Oman fragments out of the primary land-plant inventory; described the recording rate as fossilisation alone; did not show that 18 of the 28 older Murzuq zeros are younger than 459.1 Ma; and read shape diagnostics on allocated ages as rejections. All are corrected here. The analyses it requested (interval feasibility, an age–intensity frontier, shared timescale-boundary scenarios, Murzuq block dependence, influential-record checks and a public raw-input rebuild) are reported in S2.5, S5.2–S5.4 and S8 (`docs/audits/2026-11-07-stage2-handoff.md`). Two of their findings changed earlier statements: no admissible allocation rescues the 10 Myr decline, but some rescue the 25 Myr decline, which earlier drafts had listed as fitting poorly; and the canonical rebuild's executable membership test found that E4-only tracheid records had entered the polysporangiophyte category (S5.1). The 5th–95th percentiles of independent allocations for constant intensity changed from 0.073–0.765 to 0.056–0.715 when the random streams were keyed to each analysis cell; this is a new Monte Carlo sample, not a data change.

The GPT-6 Pro follow-up review (`docs/audits/2026-11-07-gpt6pro-review2-triage.md`) of commit `342153a` found that the stage-2 admission repair had removed two records from B1 because they lacked an E3 row, although E3 had never been enumerated for Ludlow and Pridoli records; both have published branched fertile material and are restored (S5.1). It also found that the Yea entry in the pivotal-record table listed sporangia that are not reported from the material claimed as Silurian; that the brief labelled a conditional bound as an earliest start, drew the oldest finds as continuous records, and compressed the spore-to-capsule

interval to "within ten million years"; that the *C. barrandei* anchor comparison mixed timescales; and that the interval search was reported on the asymptotic reference and with a blanket statement that optimisation invalidates the tests. All are corrected, and the finite-sample reference is now primary (S5.2).

Review C wording (`docs/audits/2026-11-05-review-c.md`, `docs/audits/2026-11-05-review-c-methods.md`). The previous draft called the B0–spore separation a demonstrated, observed lag; applied the $\rho f \approx 0.09$ product to both Murzuq history classes when it holds for the constant class only; reported the QSIM minimum ratios as percentages of all spores (0.307 as 31% rather than a 23.5% share); showed shape diagnostics for one age treatment only; and presented the tenfold suppression and the 15–23 Myr delays as if they were estimated. All are corrected here, and the Oman sensitivity was added.

Loop 2 corrections (`docs/loop2-close.md`, `docs/audits/2026-11-01-loop2-close-claude-response.md`) fixed a misread H0c column, an overgeneralised exponent sensitivity, an unsupported circularity claim and an overstated examination count in an earlier draft.

S8. Reproduction and navigation

From the repository root:

```
make check-public       # public tests; no copyrighted text or private outputs needed
make reproduce-derived  # selected tables; declared optimiser tolerance
make rebuild-source     # restore the archived PBDB extract and rebuild inventories and results
uv run manuscript/figures.py
python3 manuscript/tables.py
uv run tools/refs.py build
make -C manuscript
python3 manuscript/package.py
```

The body-record analysis is reproduced by `models/f3/loop3/build_reviewed.py`, `analysis.py`, `simulation.py`, `suppression_sensitivity.py` and `figures_reviewed.py`. Hash receipts bind their inputs and outputs. Frozen Loop 1 and Loop 2 runs are retained without regeneration. Third-party deposits are fetched by `data/external/scripts/deposits_fetch.py` and verified against `MANIFEST.tsv`. `manuscript/claims.csv` links each main claim to its evidence and limits, and `manuscript/value-bindings.csv` binds each quoted number of the strictly checked claims to one row and field of its evidence. That check catches swapped or drifted values; it does not check what a sentence means. The frozen PBDB all-taxa collections extract (27,830 collections) is archived under CC0 at https://doi.org/10.5281/zenodo.24028289; `make fetch-pbdb` restores it and verifies its SHA-256. `make check-public` restores/hash-checks that input before the tests. Reproduction requires byte identity except for the Murzuq optimisation CSV, whose floating-point fields have absolute tolerance 10^{-9} while its schema, row order, keys and categorical fields stay exact; receipts record both reference and regenerated SHA-256 hashes. `make check-sources` reports the public thesis route https://etheses.whiterose.ac.uk/id/eprint/4475/ and expected hash even when the copyrighted PDF is absent. The held source is the related 2013 thesis, not the 2011 abstract. Its SHA-256 is `c0e3bf798fb3c27db0bf9cbb155396be5b4b8e003ee2228defb615a65abf3211`; the source check verifies it and the byte count before checking the observation extraction. No source text is redistributed. `makedoc.lock.json` pins the document builder.

References

Abuhmida, Faisal Hadi. 2013. "Palynological Analysis of the Ordovician to Lower Silurian Sediments from the Murzuq Basin, Southwest Libya." PhD thesis, University of Sheffield. https://etheses.whiterose.ac.uk/id/eprint/4475/.

Berger, Roger L., and Dennis D. Boos. 1994. "*P* Values Maximized Over a Confidence Set for the Nuisance Parameter." *Journal of the American Statistical Association* 89 (427): 1012–16. https://doi.org/10.1080/01621459.1994.10476836.

Edwards, D., and E. C. W. Rogerson. 1979. "New Records of Fertile Rhyniophytina from the Late Silurian of Wales." *Geological Magazine* 116 (2): 93–98. https://doi.org/10.1017/s0016756800042503.

Hooper, Jack, Annabel M. Worth, Charles H. Wellman, James W. Clark, and Philip C. J. Donoghue. 2026. "Reconstructing Spore Evolution Recalibrates the Timescale of Land Plant Diversification." *Nature Plants*. https://doi.org/10.1038/s41477-026-02432-3.

Libertín, Milan, Jiří Kvaček, Jiří Bek, and Stephen McLoughlin. 2022. "The Early Land Plant Cooksonia Bohemica from the Pridoli, Late Silurian, Barrandian Area, the Czech Republic, Central Europe." *Historical Biology* 35 (12): 2504–14. https://doi.org/10.1080/08912963.2022.2144286.

Liu, Bing-Cai, Rui-Wen Zong, Kai Wang, Jiao Bai, Yi Wang, and Hong-He Xu. 2024. "Evolution of Silurian Phytogeography, with the First Report of Aberlemnia (Rhyniopsida) from the Pridoli of West Junggar, Xinjiang, China." *Palaeogeography, Palaeoclimatology, Palaeoecology* 633: 111903. https://doi.org/10.1016/j.palaeo.2023.111903.

Marshall, Charles R. 1990. "Confidence Intervals on Stratigraphic Ranges." *Paleobiology* 16 (1): 1–10. https://doi.org/10.1017/s0094837300009672.

———. 1994. "Confidence Intervals on Stratigraphic Ranges: Partial Relaxation of the Assumption of Randomly Distributed Fossil Horizons." *Paleobiology* 20 (4): 459–69. https://doi.org/10.1017/s0094837300012938.

———. 1997. "Confidence Intervals on Stratigraphic Ranges with Nonrandom Distributions of Fossil Horizons." *Paleobiology* 23 (2): 165–73. https://doi.org/10.1017/s0094837300016766.

Morris, Jennifer L., Mark N. Puttick, James W. Clark, Dianne Edwards, Paul Kenrick, Silvia Pressel, Charles H. Wellman, Ziheng Yang, Harald Schneider, and Philip C. J. Donoghue. 2018a. "Reply to Hedges Et Al.: Accurate Timetrees Do Indeed Require Accurate Calibrations." *Proceedings of the National Academy of Sciences* 115 (41). https://doi.org/10.1073/pnas.1812816115.

———. 2018b. "The Timescale of Early Land Plant Evolution." *Proceedings of the National Academy of Sciences* 115 (10). https://doi.org/10.1073/pnas.1719588115.

Rubinstein, Claudia V., and Vivi Vajda. 2019. "Baltica Cradle of Early Land Plants? Oldest Record of Trilete Spores and Diverse Cryptospore Assemblages; Evidence from Ordovician Successions of Sweden." *GFF* 141 (3): 181–90. https://doi.org/10.1080/11035897.2019.1636860.

Strauss, David, and Peter M. Sadler. 1989. "Classical Confidence Intervals and Bayesian Probability Estimates for Ends of Local Taxon Ranges." *Mathematical Geology* 21 (4): 411–27. https://doi.org/10.1007/bf00897326.

Strother, Paul K. 2016. "Systematics and Evolutionary Significance of Some New Cryptospores from the Cambrian of Eastern Tennessee, USA." *Review of Palaeobotany and Palynology* 227: 28–41. https://doi.org/10.1016/j.revpalbo.2015.10.006.

Taylor, Wilson A., and Paul K. Strother. 2024. "Ultrastructure of Cambrian Cryptospores and the Early Evolution of the Plant Spore Wall." *Fossil Imprint* 80 (1): 90–106. https://doi.org/10.37520/fi.2024.009.

Vecoli, Marco, Charles H. Wellman, Philippe Gerrienne, Alain Le Hérissé, and Philippe Steemans. 2017. "Middle Ordovician Cryptospores from the Saq-Hanadir Transitional Beds in the QSIM-801 Well, Saudi Arabia." *Revue de Micropaléontologie* 60 (3): 319–31. https://doi.org/10.1016/j.revmic.2017.06.003.

Wang, Kai, Hong-He Xu, and Lei-Ming Yin. 2022. "A Palynological Assemblage from the Cambrian (Series 2, Stage 4) of Shandong Province, China, and Its Implications to the Transition from Algae to Land Plants." *Review of Palaeobotany and Palynology* 301: 104645. https://doi.org/10.1016/j.revpalbo.2022.104645.

Wang, Steve C., Philip J. Everson, Heather Jianan Zhou, Dasol Park, and David J. Chudzicki. 2016. "Adaptive Credible Intervals on Stratigraphic Ranges When Recovery Potential Is Unknown." *Paleobiology* 42 (2): 240–56. https://doi.org/10.1017/pab.2015.37.

Wellman, Charles H., Peter L. Osterloff, and Uzma Mohiuddin. 2003. "Fragments of the Earliest Land Plants." *Nature* 425 (6955): 282–85. https://doi.org/10.1038/nature01884.

Supplementary Information

Fig. S1.

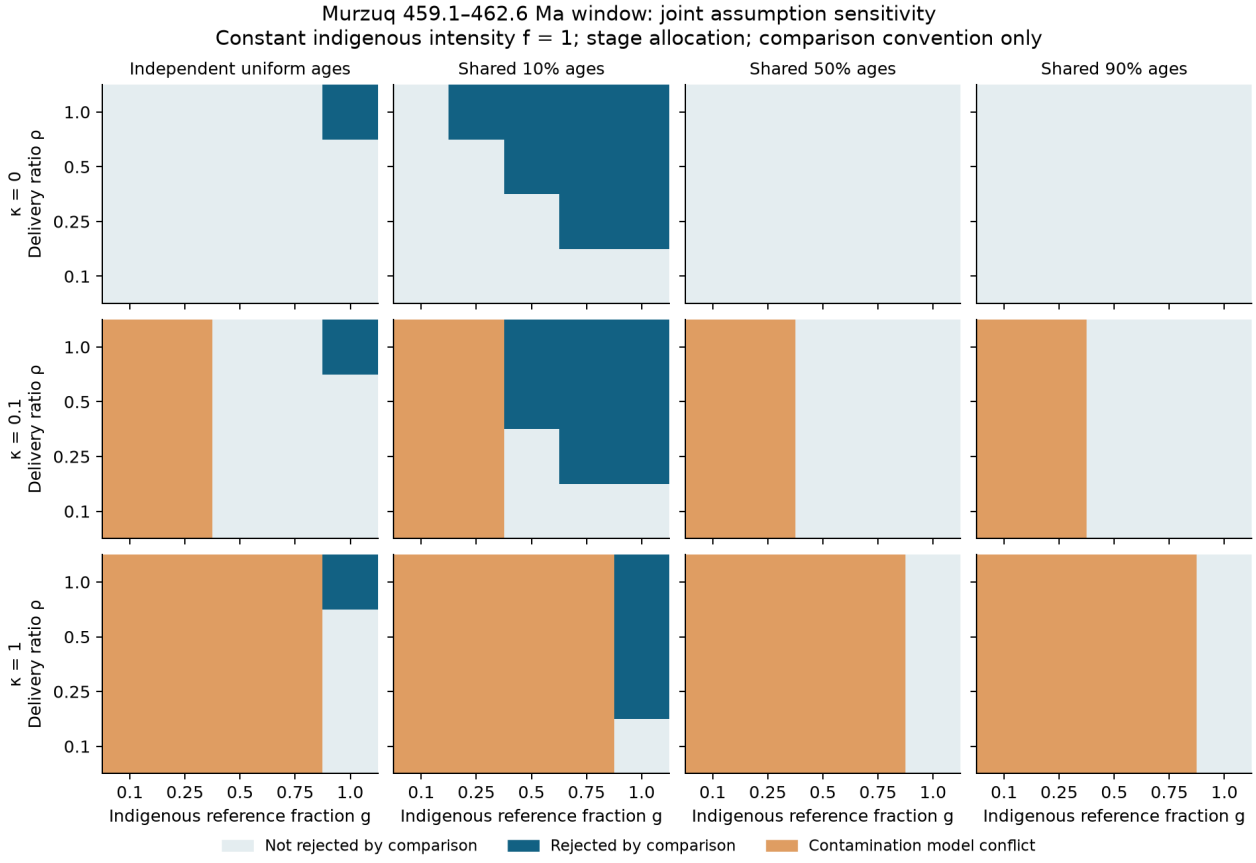


Figure 1: Joint sensitivity of the short Murzuq window. Colours report an uncalibrated comparison or model-conflict flag, not confidence or probabilities over geology.

Fig. S2.

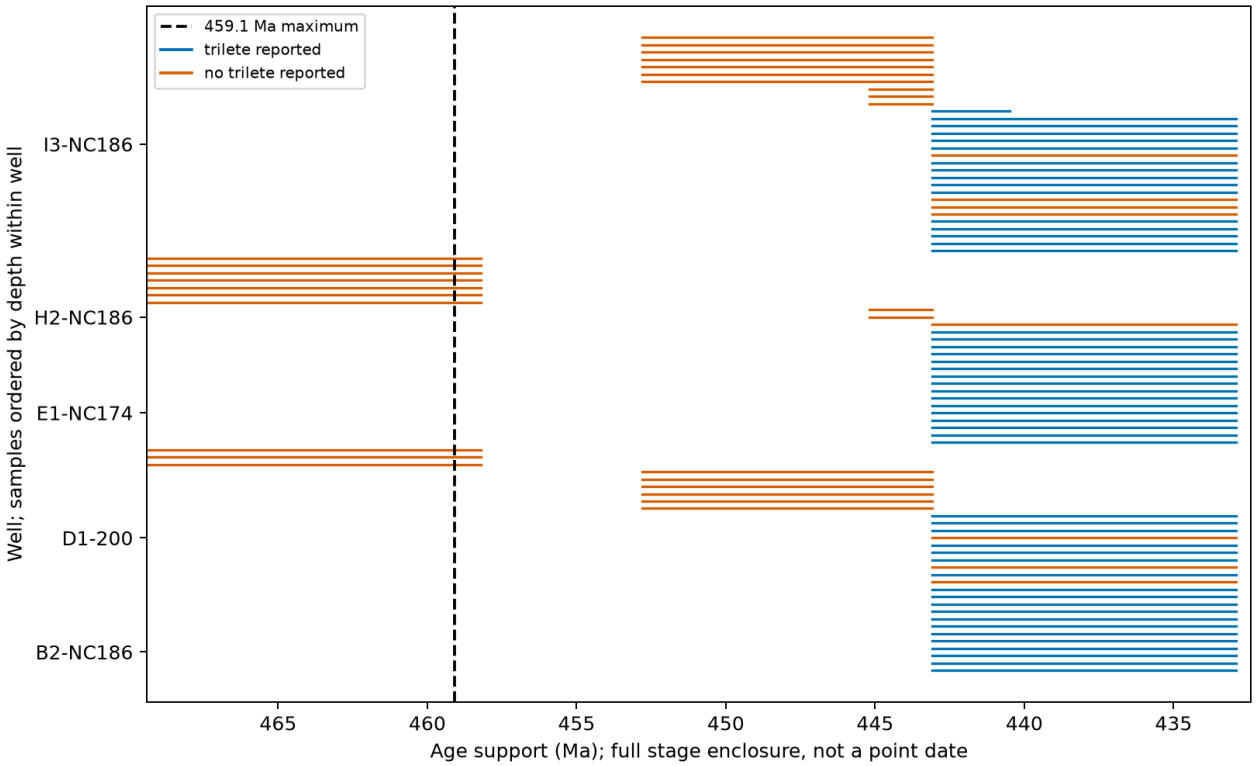


Figure 2: Murzuq Basin cuttings by well, ordered by depth: each line is a cutting's full stage enclosure, not a point date. Ten Darriwilian cuttings in two wells have possible support older than the 459.1 Ma maximum (dashed); the other 18 older cuttings are wholly younger. Generated by `models/f3/loop3/murzuq_stage2.py`.

Fig. S3.

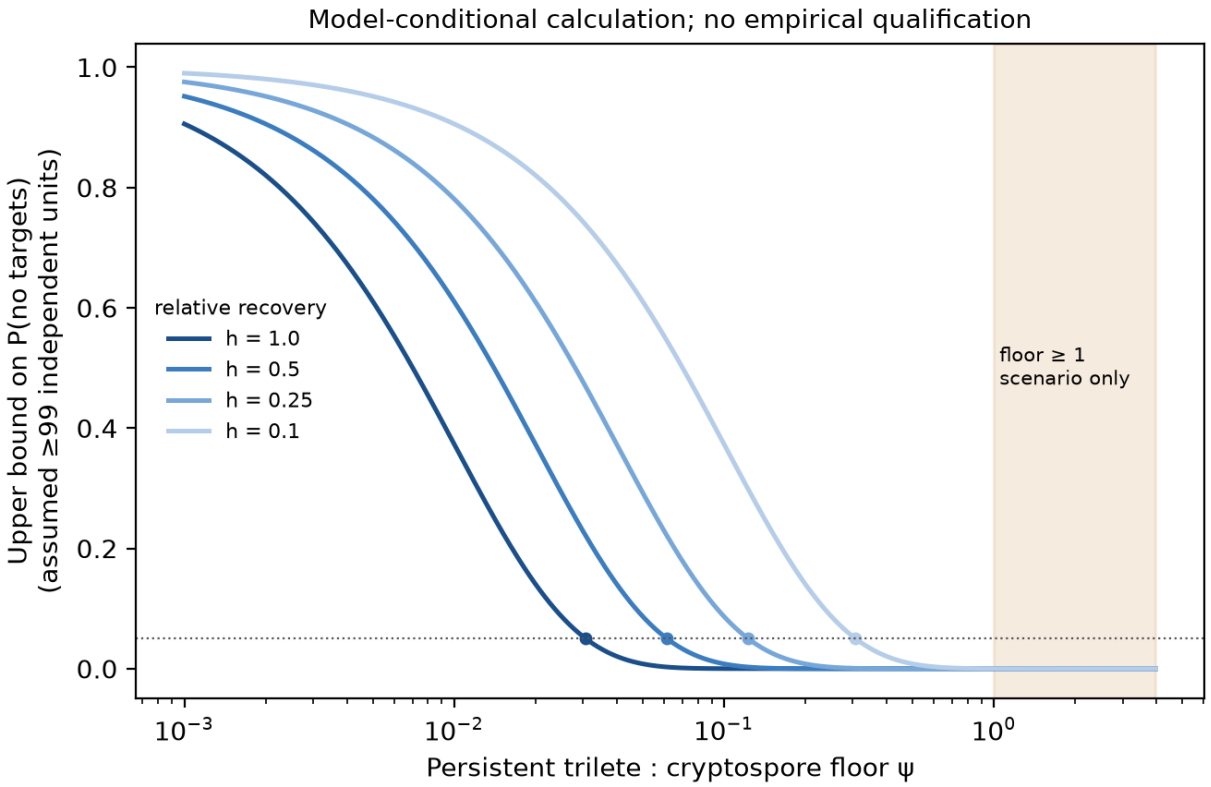


Figure 3: Upper bound on the all-zero probability under the hypothetical independent, co-ascertained unit model; points mark the 0.05 threshold.

Fig. S4.

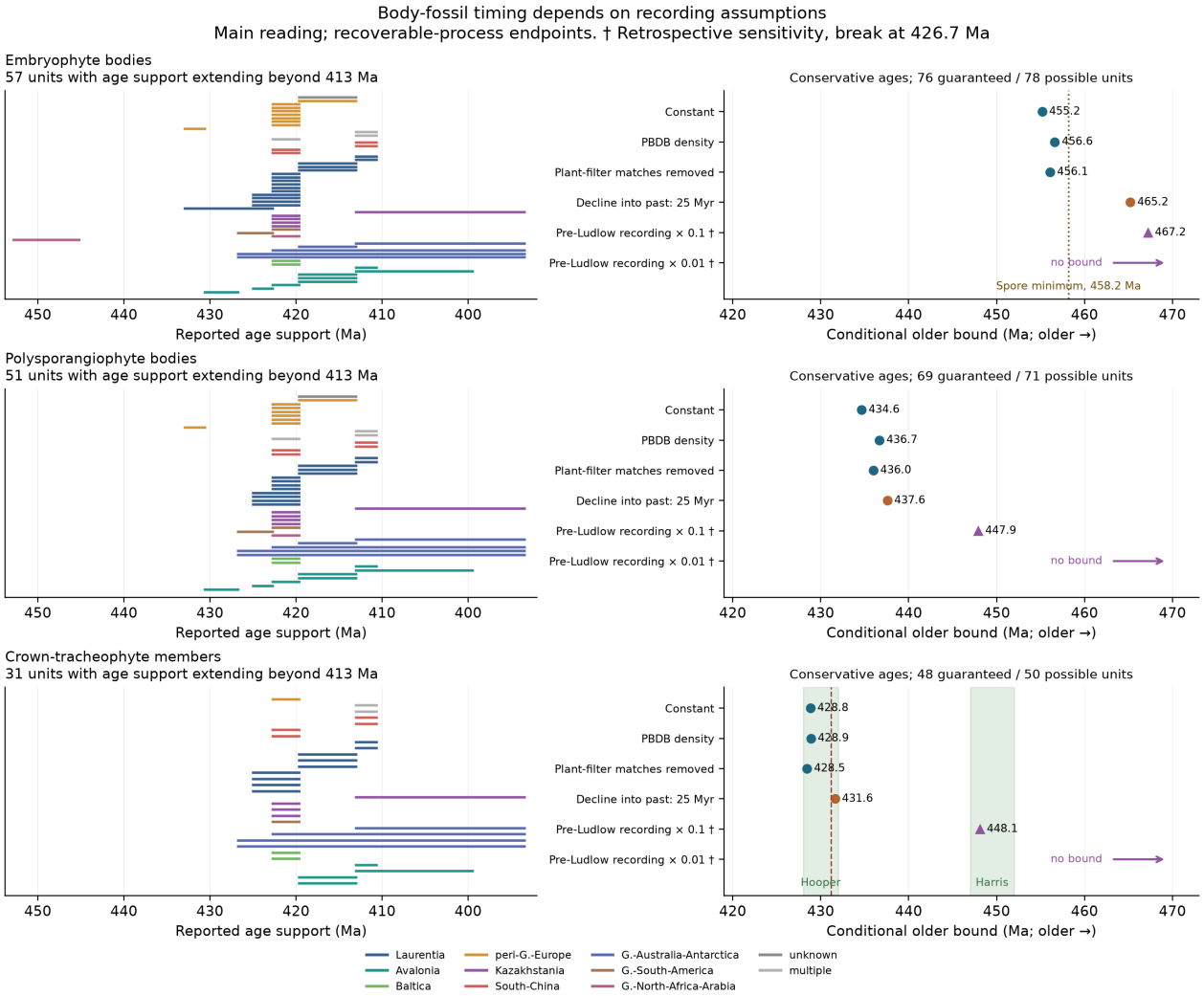


Figure 4: Reviewed main macrofossil figure as delivered by the analysis (`models/f3/loop3/out/fig_macro_recovery.pdf`).

Fig. S5.

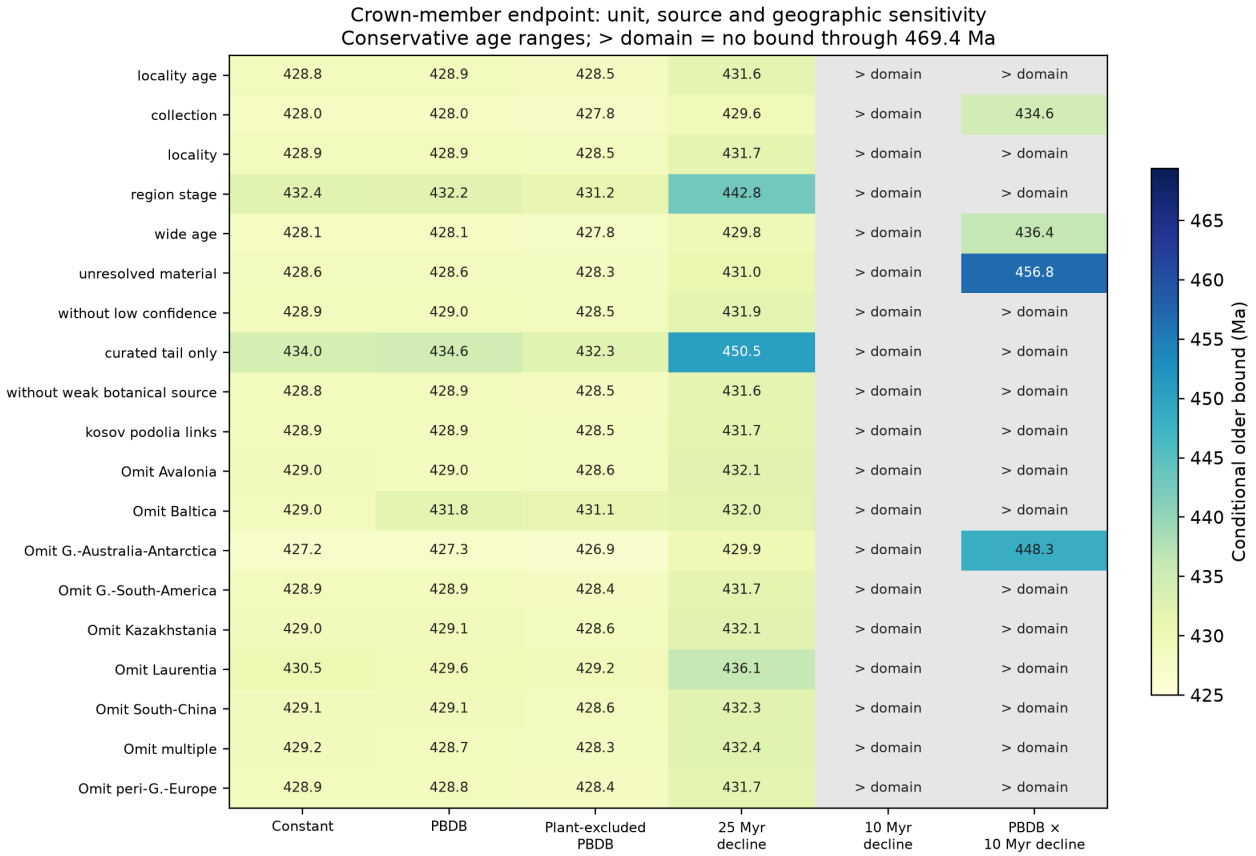


Figure 5: Sensitivity of the crown vascular-plant bound to units, sources, regions and age rules.

Fig. S6.

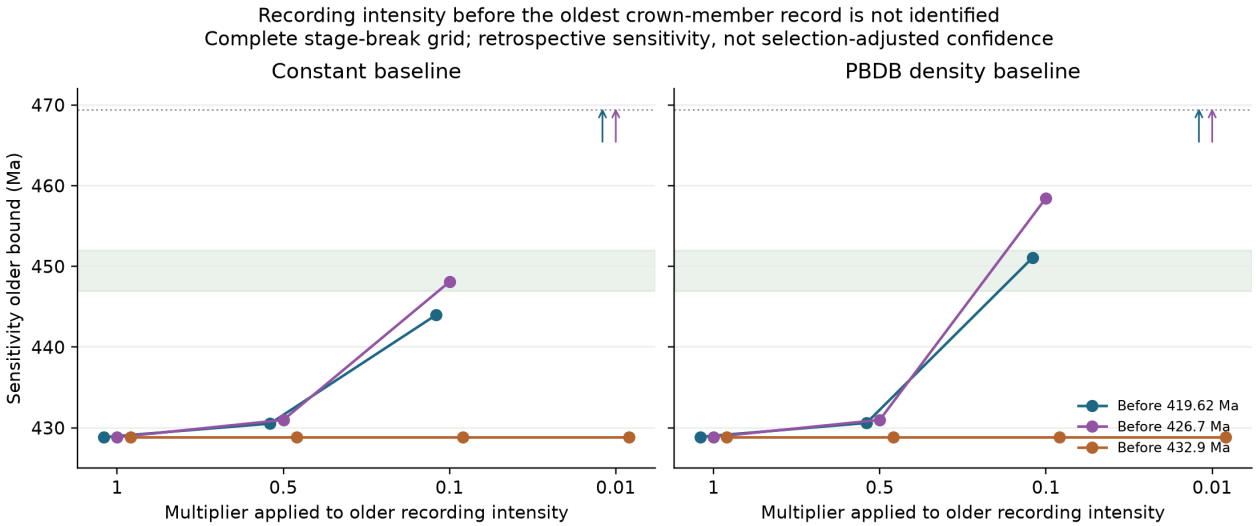


Figure 6: Retrospective recovery-suppression grid.

Fig. S7.

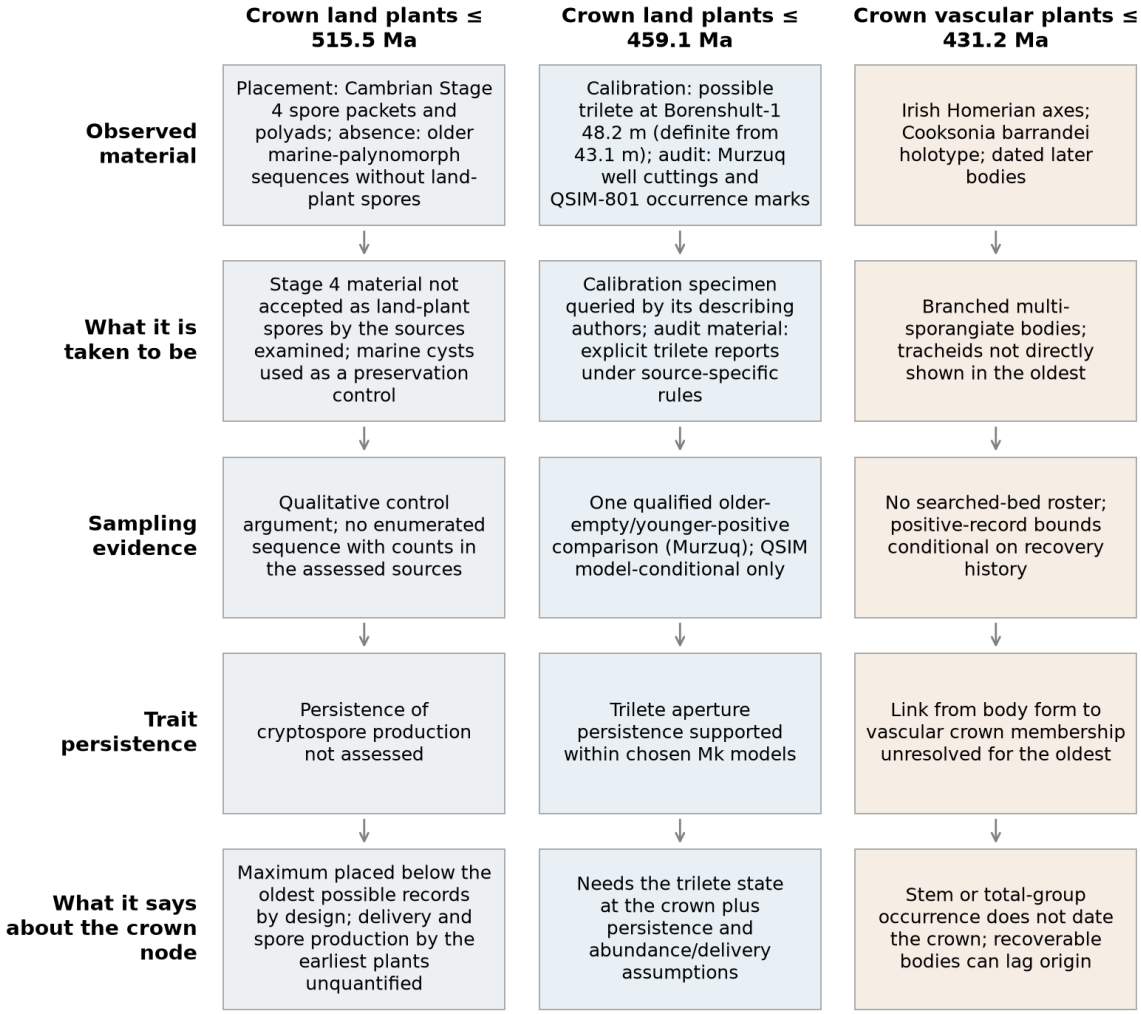


Figure 7: From observed material to node interpretation. For each maximum, the observed material, what it is taken to be, the sampling evidence, the persistence of the diagnostic trait and the implication for the crown node are separate links; a maximum needs all five. `manuscript/figures.py`.